

Tuyet Minh Vu

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EDUCATION

University of Utah – John & Marcia Price College of Engineering

Salt Lake City, UT

Bachelor of Science in Computer Science

Expected May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Practice, Machine Learning

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, JavaScript, HTML/CSS

Frameworks & Libraries: PyTorch, NumPy, Scikit-learn, Blazor, React

Tools & Technologies: Git/GitHub, GitLab, Unity, SQL, Figma, Qt

RELEVANT EXPERIENCE

University of Utah – Therapeutic Games and Apps Lab

Salt Lake City, UT

Undergraduate Software Developer / Research Assistant

May 2026 – July 2026

- Maintained Color Me Healthy, a Unity WebGL pediatric health game used in active cancer treatment settings, supporting its migration from a standalone Android application to a cross-platform WebGL client.
- Identified, reproduced, and resolved 10+ platform-specific bugs introduced by the Android-to-WebGL migration, spanning frontend functionality, UI/UX, and game asset consistency.
- Diagnosed defects using network inspection, log tracing, and breakpoint debugging, following a structured reproduce-isolate-diagnose-fix-validate workflow, and tracked fixes through a project Kanban board.
- Conducted cross-platform testing across multiple devices and browsers in coordination with a clinical team to validate gameplay stability and symptom-tracking accuracy before deployment.

PROJECTS

Learning Management System | C#, SQL, Blazor

April 2026

- Architected a normalized relational database with constraints and indexing to enforce data integrity and support scalable multi-table queries across courses, users, and assignments.
- Implemented automated grading logic to dynamically compute student scores from submissions, assignment categories, and configurable weights, eliminating manual calculation.
- Integrated a Blazor front-end with SQL backend to surface real-time grade updates, enrollment data, and course dashboards with clean component-level data binding.

Cooking Educational Game | C++, Qt, Box2D

April 2026

- Built a physics-based cooking education game in C++ with Qt and Box2D that teaches recipe concepts through real-time ingredient simulations, delivering a polished end-to-end experience.
- Engineered live physics using Box2D forces and frame-by-frame motion updates to simulate pouring, mixing, and whisking with distinct particle effects for dry and liquid ingredients.
- Designed a custom cursor interaction system with timer-based animations for ingredient selection, increasing engagement through tactile, responsive feedback.

Real-Time Multiplayer Snake Game | C#, Blazor, SQL, TCP Networking

November 2025

- Architected a real-time multiplayer client in C# using asynchronous TCP networking and an event-driven model to handle concurrent player connections with minimal latency.
- Built a dynamic Blazor UI rendering snakes, walls, and power-ups in real time with low-latency keyboard input keeping player state synchronized with the server.
- Optimized rendering by deserializing JSON server broadcasts and triggering targeted component redraws, reducing unnecessary re-renders and improving frame consistency.
- Integrated a SQL backend via REST endpoints to persist per-game and per-player statistics, powering a live leaderboard and post-game analytics dashboard.

LEADERSHIP EXPERIENCE

Salt Lake Community College – Programming Club

Salt Lake City, UT

Officer

September 2024 – May 2025

- Designed and led weekly LeetCode-style coding challenges in data structures and algorithms for 200+ students, improving technical interview readiness across the club.
- Authored problem sets with edge cases, test suites, and solution walkthroughs covering time and space complexity to build rigorous algorithmic thinking.
- Grew club engagement through targeted social media campaigns, classroom outreach, and coordination of coding competitions and technical workshops.